

with provisions for the factors of safety and degradation of the chemistry over the expected life of customised battery solutions."

Today, original equipment manufacturers (OEMs) demand batteries with highly customised footprints and performance characteristics, which is impossible with batteries like nickel-metal hydride (NiMH) and nickel-cadmium (NiCd).

"OEMs demand an X ampere-hour battery to provide peak pulse currents of 18X-30X amperes in order to meet conditional operations of the device, which is definitely impossible with legacy chemistries. Light weight, high energy density, safe materials, fast recharging, high continuous current and peak pulse from small-capacity cells, and consistency in quality are the requirements of OEMs today," adds Ali.

Li-ion batteries fit the bill

It is the battery and its chemistry that make a product energy-efficient. Li-ion batteries are environment-friendly compared to NiCd and sealed lead-acid (SLA) batteries which were commonly used earlier. Basically, these are secondary (rechargeable) batteries that use the electrochemical reaction between a negative lithium anode and a cathode to generate an electric current. Most rechargeable (able to reverse the electrochemical reaction in the opposite direction of discharge) batteries are also easily recyclable.

You might have heard that repeated charging of your cellphone or laptop degrades its performance and life. Well, it was true until Li-ion were used in these devices. The performance and shelf life of batteries degrade due to 'memory effect.' When a battery is recharged before it has discharged half the power, it tends to forget its original power capacity. As cadmium crystals are the culprits, NiMH and Li-ion batteries have no memory effect.

Lithium batteries show a high volumetric energy density. Ali recounts, "Polymer lithium (PLI) batteries have 3.7V per cell, which is thrice the volt-



Li-ion battery for mobile phones

age of NiCd or NiMH. Further, these are very long-lasting, robust and maintenance-free. They can be built into a rugged assembly battery to suit the requirement."

"It is because of the Lithium substance that a Li-ion battery has a higher energy density in a smaller size and

weight than previous technologies," explains Shah. They also have lower self-discharge rate—typically, only 5 per cent loss over a month—which means longer shelf life.

As regards the cost, Kandhari cites, "When it comes to battery pack, Li-ion batteries are the most cost-effective. For applications which demand higher voltage and current ratings, many batteries are packed together. For example, if the demand is 6V, you will need five 1.2V NiMH/NiCd batteries. The cost increases with the number of batteries. In the case of Li-ion, you need two 3.7V batteries to give you 7.4V. Thus the cost and size both are affected." Also, the prices of nickel metal are increasing day-by-day.

Understanding their basics

A lot of people confuse between lithium and Li-ion batteries. These differ in that a lithium battery uses lithium in pure form, which raises its energy density considerably but makes it highly unstable too, while a Li-ion battery uses lithium compounds, which makes it much more stable but pack lesser energy density than lithium batteries.

Lithium-ion: Indian scenario

There are nearly 10 million mobile devices sold in India, making India the largest consumer of mobile phones. This indirectly makes India a big consumer of Li-ion batteries too.

However, Laxman Soma, CEO, GSK Technologies, informs, "For the Rs 600-million business of mobile phones, every battery used is imported." There is no Li-ion battery manufacturing being done in India. Soma suggest that if we really want to promote the use of hybrid electric vehicles, it's necessary to have a Li-ion battery manufacturing set-up in India.

At one of the forums, Sharon Xiao, vice president, BAK, said, "Production of Li-ion batteries is a very difficult and complex process. It needs a very clean environment for manufacturing as any kind of contamination may affect the performance of the battery."

In the recent past, we have experienced a lot of low-quality or duplicate Li-ion batteries being supplied which greatly harm the electronics. To facilitate the promotion of standards, use of original batteries and positive future outlook, India and China Lithium-ion Technology Forum was formed. The forum will primarily focus on technical specifications for mobile phones and Li-ion batteries.

Comparing situation in India and China, Soma says, "The Chinese government has provided funding and fiscal incentives to Li-ion producers. It has also raised export tax rebates for Li-ion products. There is no funding or incentives available to the producers in India. The Indian government has to understand that there is a need for home-grown products to generate more business avenues."